



Security Operations Centre Analyst

QP Code: SSC/Q0909

Version: 4.0

NSQF Level: 5

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Qualification Pack

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SSC/Q0909: Security Operations Centre Analyst

Brief Job Description

The Analyst in the Security Operations Centre is responsible for detecting threats and monitoring events using Sysinternals tools. They analyze security events, respond to detected threats, and identify and mitigate cyber-attacks, ensuring robust protection of information systems while collaborating effectively within the security team.

Personal Attributes

This job may require the individual to work in a team/shifts. The individual should be result oriented and have a high attention for detail. The individual should also be able to demonstrate good communication skills and logical thinking with willingness to work in shifts.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N0961: Detect threats and monitor events using Sysinternal Tools](#)
2. [SSC/N0962: Analyze security events and response to detected threats](#)
3. [SSC/N0963: Identify and response to cyber attack](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Information and Cyber Security
Country	India
NSQF Level	5
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL

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Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (of 3-year/ 4-year UG) with NA of experience OR Completed 3 year diploma after 10th with 1.5 years of experience in relevant field OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience in relevant field
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	4.0
Reference code on NQR	QG-05-IT-03638-2025-V2-NASSCOM
NQR Version	4.0

Qualification Pack

SSC/N0961: Detect threats and monitor events using Sysinternal Tools

Description

The OS unit is about using Sysinternal Tools to detect security threats and monitor system events, ensuring timely identification and response to potential risks.

Scope

The scope covers the following :

- Monitor and Analyze Security Threats in ICT Systems
- Threat Monitoring and Incident Analysis
- Threat Detection and Log Analysis for Cybersecurity Operations
- Security Operations Monitoring and Incident Management

Elements and Performance Criteria

Monitor and Analyze Security Threats in ICT Systems

To be competent, the user/individual on the job must be able to:

- PC1.** confirm the range of information assets and system components to be monitored in collaboration with authorized personnel
- PC2.** employs designated monitoring and data collection techniques and tools in accordance with organizational procedures and policies
- PC3.** oversee the organization's traffic and logs generated by ICT systems using various security technologies to identify security threats and assess the health of the ICT systems
- PC4.** oversee external data sources such as Computer Network Defense (CND) vendor websites, Computer Emergency Response Teams, SANS, and Security Focus
- PC5.** identify security issues that could affect the enterprise
- PC6.** perform detailed email analysis to identify phishing attempts by reviewing suspicious elements such as email addresses, links, or domain names, ensuring alignment with legitimate sources
- PC7.** analyze unexpected attachments or links in emails to detect and prevent potential malware downloads, utilizing phishing detection techniques to safeguard the organization's network
- PC8.** verify sender authenticity by cross-checking the sender's email address against the organization it claims to represent, applying phishing analysis methods to prevent fraudulent emails from compromising security

Threat Monitoring and Incident Analysis

To be competent, the user/individual on the job must be able to:

- PC9.** use Sysinternals tools (such as Process Explorer, Autoruns, and TCPView) to identify and examine potential malware or suspicious processes on the network
- PC10.** utilize Sysinternal tools (such as Process Explorer, Autoruns, and TCPView) for thorough system investigations to detect unauthorized changes or applications that may impact security.
- PC11.** gather logs from all types of ICT systems, devices, and applications as mandated by the organization

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- PC12.** maintain continuous monitoring of all critical infrastructure, including servers, cloud workloads, applications, and networks, to identify signs of intrusion or suspicious activity around the clock
- PC13.** employ SIEM (Security Information and Event Management) or XDR (Extended Detection and Response) tools to consolidate alerts, examine telemetry data, and identify threats in real-time
- PC14.** analyze trends and patterns utilizing the SIEM tool
- PC15.** collaborate with enterprise-wide Computer Network Defense (CND) teams to verify network alerts
- PC16.** conduct event correlation using collected information to enhance situational awareness and assess the potential threat level

Threat Detection and Log Analysis for Cybersecurity Operations

To be competent, the user/individual on the job must be able to:

- PC17.** conduct a thorough analysis of log data gathered from different network events to identify anomalies
- PC18.** recognize standard activities and detect anomalies that suggest suspicious behavior
- PC19.** support the identification of real cyber threats by filtering out false positives and classifying threats based on their severity to ensure prompt action
- PC20.** utilize AI-powered SIEM tools to enhance the precision of threat detection progressively.
- PC21.** prioritize the service request according to organizational procedures and policies
- PC22.** obtain help or advice from specialist if the problem is outside his/her area of competence or experience

Security Operations Monitoring and Incident Management

To be competent, the user/individual on the job must be able to:

- PC23.** report the results of the monitoring, ticket raising and ticket closure activities using standard documentation following organisational procedures
- PC24.** participate in 24/7 security operations centre shift schedule
- PC25.** receive shift handover along with relevant information, authorities and instructions
- PC26.** comply with relevant legislation, standards, policies and procedures
- PC27.** maintain a knowledge base of the known problems
- PC28.** use escalation matrix for unresolved tickets within agreed turnaround times

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The principles of Security Information and Event Management (SIEM) and Extended Detection and Response (XDR) tools, including their role in consolidating alerts and identifying threats in real-time.
- KU2.** Various cyber threats, including phishing, malware, ransomware, and insider threats, and the methodologies used to detect, analyze, and mitigate them.
- KU3.** Log analysis techniques and how to gather, interpret, and correlate logs from different ICT systems, applications, and network events to detect anomalies.

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- KU4.** The fundamentals of Computer Network Defense (CND), including the importance of external threat intelligence sources such as CERT, SANS, and Security Focus.
- KU5.** The importance of AI-powered security tools in enhancing the accuracy of threat detection and how they improve over time through continuous learning.
- KU6.** The organizational policies, regulations, and compliance frameworks (such as GDPR, NIST, and ISO 27001) that govern cybersecurity monitoring and incident response.
- KU7.** The processes and best practices for conducting email security analysis, including identifying phishing attempts, verifying sender authenticity, and analyzing malicious attachments.
- KU8.** The procedures for security operations center (SOC) shift management, including shift handover protocols, documentation standards, and escalation matrix guidelines.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Utilize SIEM, XDR, and threat intelligence platforms to conduct real-time monitoring and threat detection, filtering out false positives while identifying real security threats.
- GS2.** Analyze logs and telemetry data to detect abnormal patterns, correlate security events, and assess the level of potential cyber threats to the organization.
- GS3.** Investigate phishing emails by analyzing links, domain names, and attachments, verifying sender authenticity, and implementing measures to prevent fraudulent email attacks.
- GS4.** Use Sysinternals tools (such as Process Explorer, Autoruns, and TCPView) to identify suspicious processes, unauthorized applications, and security vulnerabilities in ICT systems.
- GS5.** Prioritize and classify cybersecurity incidents based on severity, urgency, and business impact, ensuring timely response and mitigation actions.
- GS6.** Follow established escalation protocols to report security incidents, raise service requests, and allocate tickets to the appropriate teams for resolution.
- GS7.** Maintain detailed documentation of security monitoring activities, including incident reports, ticket closure records, and known threat intelligence updates.
- GS8.** Collaborate effectively with enterprise-wide CND teams, IT personnel, and security specialists to ensure a coordinated response to detected threats and cyber incidents.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Monitor and Analyze Security Threats in ICT Systems</i>	9	12	-	6
PC1. confirm the range of information assets and system components to be monitored in collaboration with authorized personnel	1	2	-	1
PC2. employs designated monitoring and data collection techniques and tools in accordance with organizational procedures and policies	2	2	-	1
PC3. oversee the organization's traffic and logs generated by ICT systems using various security technologies to identify security threats and assess the health of the ICT systems	1	2	-	1
PC4. oversee external data sources such as Computer Network Defense (CND) vendor websites, Computer Emergency Response Teams, SANS, and Security Focus	1	2	-	-
PC5. identify security issues that could affect the enterprise	1	1	-	1
PC6. perform detailed email analysis to identify phishing attempts by reviewing suspicious elements such as email addresses, links, or domain names, ensuring alignment with legitimate sources	1	1	-	1
PC7. analyze unexpected attachments or links in emails to detect and prevent potential malware downloads, utilizing phishing detection techniques to safeguard the organization's network	1	1	-	1
PC8. verify sender authenticity by cross-checking the sender's email address against the organization it claims to represent, applying phishing analysis methods to prevent fraudulent emails from compromising security	1	1	-	-
<i>Threat Monitoring and Incident Analysis</i>	8	16	-	4
PC9. use Sysinternals tools (such as Process Explorer, Autoruns, and TCPView) to identify and examine potential malware or suspicious processes on the network	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. utilize Sysinternal tools (such as Process Explorer, Autoruns, and TCPView) for thorough system investigations to detect unauthorized changes or applications that may impact security.	1	2	-	-
PC11. gather logs from all types of ICT systems, devices, and applications as mandated by the organization	1	2	-	-
PC12. maintain continuous monitoring of all critical infrastructure, including servers, cloud workloads, applications, and networks, to identify signs of intrusion or suspicious activity around the clock	1	2	-	-
PC13. employ SIEM (Security Information and Event Management) or XDR (Extended Detection and Response) tools to consolidate alerts, examine telemetry data, and identify threats in real-time	1	2	-	1
PC14. analyze trends and patterns utilizing the SIEM tool	1	2	-	-
PC15. collaborate with enterprise-wide Computer Network Defense (CND) teams to verify network alerts	1	2	-	1
PC16. conduct event correlation using collected information to enhance situational awareness and assess the potential threat level	1	2	-	1
<i>Threat Detection and Log Analysis for Cybersecurity Operations</i>	7	12	-	6
PC17. conduct a thorough analysis of log data gathered from different network events to identify anomalies	2	2	-	1
PC18. recognize standard activities and detect anomalies that suggest suspicious behavior	1	2	-	1
PC19. support the identification of real cyber threats by filtering out false positives and classifying threats based on their severity to ensure prompt action	1	2	-	1
PC20. utilize AI-powered SIEM tools to enhance the precision of threat detection progressively.	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. prioritize the service request according to organizational procedures and policies	1	2	-	1
PC22. obtain help or advice from specialist if the problem is outside his/her area of competence or experience	1	2	-	1
<i>Security Operations Monitoring and Incident Management</i>	6	10	-	4
PC23. report the results of the monitoring, ticket raising and ticket closure activities using standard documentation following organisational procedures	1	2	-	-
PC24. participate in 24/7 security operations centre shift schedule	1	2	-	1
PC25. receive shift handover along with relevant information, authorities and instructions	1	2	-	1
PC26. comply with relevant legislation, standards, policies and procedures	1	1	-	1
PC27. maintain a knowledge base of the known problems	1	2	-	-
PC28. use escalation matrix for unresolved tickets within agreed turnaround times	1	1	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0961
NOS Name	Detect threats and monitor events using Sysinternal Tools
Sector	IT-ITeS
Sub-Sector	
Occupation	Information and Cyber Security
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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SSC/N0962: Analyze security events and response to detected threats

Description

The OS unit is about analyzing security events and responding to detected threats promptly to mitigate risks and ensure the protection of digital assets.

Scope

The scope covers the following :

- Security Monitoring and Vulnerability Management
- Cyber Threat Detection and Incident Response
- Incident Response and Security Monitoring

Elements and Performance Criteria

Security Monitoring and Vulnerability Management

To be competent, the user/individual on the job must be able to:

- PC1.** receive and evaluate alarms and alerts from multiple sources across the enterprise, identifying potential causes and determining the underlying issues behind the alerts
- PC2.** maintain up-to-date inventories of all IT assets, including applications, databases, servers, cloud services, and endpoints, to ensure comprehensive protection and security
- PC3.** utilize asset discovery tools (Cacti, IP Scanning, etc.) to monitor and track changes, identify vulnerabilities, and assess potential security risks within the IT infrastructure
- PC4.** regularly implement patches, upgrades, and updates for security tools, including firewalls, antivirus, and monitoring software, to ensure continuous protection and optimal functioning of the Security Operations Centre (SOC)
- PC5.** contribute to the development of backup policies and procedures, ensuring data availability and integrity to support business continuity and disaster recovery efforts
- PC6.** conduct vulnerability assessments to identify potential weaknesses across IT resources, ensuring comprehensive analysis and reporting to enhance the organization's security posture
- PC7.** determine which information assets and system components could be affected by identified security incidents
- PC8.** analyze detected malicious activities to assess exploited vulnerabilities, methods of exploitation, and their impact on systems and data

Cyber Threat Detection and Incident Response

To be competent, the user/individual on the job must be able to:

- PC9.** analyze log files from multiple sources to detect potential threats to network security
- PC10.** conduct triage for computer network defense (CND) incidents, which involves evaluating the scope, urgency, and potential impact of each incident
- PC11.** conduct and assist in penetration testing activities to simulate cyberattacks, identifying vulnerabilities and strengthening the organization's security posture

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- PC12.** utilize a Security Information and Event Management (SIEM) tool to correlate and analyze events for the detection of IT security incidents
- PC13.** collect and secure evidence related to identified incidents
- PC14.** investigate the methods used to gain access to the compromised information assets and system components
- PC15.** mitigate various types of vulnerabilities and their corresponding attacks on the system
- PC16.** decide on the suitable response based on the identification and analysis of anomalous activity

Incident Response and Security Monitoring

To be competent, the user/individual on the job must be able to:

- PC17.** provide suggestions for specific actions to address incidents
- PC18.** conduct a health assessment of the security solution
- PC19.** utilize external information resources for conducting incident investigations
- PC20.** notify the appropriate individuals about any incidents that cannot be resolved or mitigated, adhering to organizational procedures
- PC21.** adhere to organizational protocols for closing incidents
- PC22.** document and report incident management activities in accordance with organizational procedures and standard documentation practices
- PC23.** monitor and record incidents from the initial detection to the final resolution using a SIEM tool
- PC24.** incorporate the assets into the SIEM solution for log analysis

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The principles of Security Information and Event Management (SIEM) tools and their role in threat detection, log correlation, and incident response.
- KU2.** Various asset discovery tools such as Cacti and IP Scanning, and how they help in identifying vulnerabilities and monitoring changes in IT infrastructure.
- KU3.** The process of penetration testing, vulnerability assessments, and different techniques used to simulate cyberattacks to identify system weaknesses.
- KU4.** The methodologies for analyzing security alerts, logs, and malicious activities to determine the root cause, methods of exploitation, and the potential impact on information assets.
- KU5.** Organizational incident management protocols, including classification, prioritization, escalation, and closure of security incidents in compliance with standard operating procedures (SOPs).
- KU6.** Techniques for collecting, preserving, and securing digital evidence related to security incidents, ensuring compliance with organizational and legal requirements.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Use SIEM tools effectively to correlate and analyze security events, detect IT security incidents, and generate reports for further action.
- GS2.** Conduct triage for security incidents, assessing the severity, scope, and urgency of threats to prioritize an appropriate response.
- GS3.** Analyze log files from various sources and identify malicious activities, suspicious patterns, and unauthorized access attempts in IT systems.
- GS4.** Implement mitigation strategies for different types of vulnerabilities and cyberattacks, ensuring quick and effective threat resolution.
- GS5.** Document, report, and communicate findings from vulnerability assessments, penetration testing, and incident response activities in a structured and compliant manner.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Security Monitoring and Vulnerability Management</i>	10	16	-	7
PC1. receive and evaluate alarms and alerts from multiple sources across the enterprise, identifying potential causes and determining the underlying issues behind the alerts	2	2	-	1
PC2. maintain up-to-date inventories of all IT assets, including applications, databases, servers, cloud services, and endpoints, to ensure comprehensive protection and security	1	2	-	1
PC3. utilize asset discovery tools (Cacti, IP Scanning, etc.) to monitor and track changes, identify vulnerabilities, and assess potential security risks within the IT infrastructure	1	2	-	1
PC4. regularly implement patches, upgrades, and updates for security tools, including firewalls, antivirus, and monitoring software, to ensure continuous protection and optimal functioning of the Security Operations Centre (SOC)	2	2	-	1
PC5. contribute to the development of backup policies and procedures, ensuring data availability and integrity to support business continuity and disaster recovery efforts	1	2	-	1
PC6. conduct vulnerability assessments to identify potential weaknesses across IT resources, ensuring comprehensive analysis and reporting to enhance the organization's security posture	1	2	-	1
PC7. determine which information assets and system components could be affected by identified security incidents	1	2	-	-
PC8. analyze detected malicious activities to assess exploited vulnerabilities, methods of exploitation, and their impact on systems and data	1	2	-	1
<i>Cyber Threat Detection and Incident Response</i>	11	18	-	8
PC9. analyze log files from multiple sources to detect potential threats to network security	1	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. conduct triage for computer network defense (CND) incidents, which involves evaluating the scope, urgency, and potential impact of each incident	1	2	-	1
PC11. conduct and assist in penetration testing activities to simulate cyberattacks, identifying vulnerabilities and strengthening the organization's security posture	2	2	-	1
PC12. utilize a Security Information and Event Management (SIEM) tool to correlate and analyze events for the detection of IT security incidents	2	2	-	1
PC13. collect and secure evidence related to identified incidents	1	2	-	1
PC14. investigate the methods used to gain access to the compromised information assets and system components	2	3	-	1
PC15. mitigate various types of vulnerabilities and their corresponding attacks on the system	1	2	-	1
PC16. decide on the suitable response based on the identification and analysis of anomalous activity	1	2	-	1
<i>Incident Response and Security Monitoring</i>	9	16	-	5
PC17. provide suggestions for specific actions to address incidents	1	2	-	1
PC18. conduct a health assessment of the security solution	1	2	-	1
PC19. utilize external information resources for conducting incident investigations	1	2	-	-
PC20. notify the appropriate individuals about any incidents that cannot be resolved or mitigated, adhering to organizational procedures	1	2	-	1
PC21. adhere to organizational protocols for closing incidents	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. document and report incident management activities in accordance with organizational procedures and standard documentation practices	1	2	-	1
PC23. monitor and record incidents from the initial detection to the final resolution using a SIEM tool	2	2	-	-
PC24. incorporate the assets into the SIEM solution for log analysis	1	2	-	-
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0962
NOS Name	Analyze security events and response to detected threats
Sector	IT-ITeS
Sub-Sector	
Occupation	Information and Cyber Security
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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SSC/N0963: Identify and response to cyber attack

Description

The OS unit is about detecting potential cyber threats and responding promptly to mitigate risks and ensure system security.

Scope

The scope covers the following :

- Threat Detection, Incident Analysis, and Response
- Incident Containment, Mitigation, and Recovery

Elements and Performance Criteria

Threat Detection, Incident Analysis, and Response

To be competent, the user/individual on the job must be able to:

- PC1.** utilize advanced security tools (Wireshark, KisMAC, NetStumbler, etc.) to monitor network traffic, system logs, and user behavior continuously for signs of suspicious activity
- PC2.** leverage threat intelligence feeds to remain informed about emerging threats and attack patterns
- PC3.** employ Security Information and Event Management (SIEM) solutions to aggregate and correlate security events from multiple sources to identify potential incidents
- PC4.** apply algorithms for detecting deviations from normal behavior, including unusual network traffic or unauthorized access attempts
- PC5.** correlate alerts from various sources when suspicious activities are detected to determine potential incidents
- PC6.** categorize incidents based on severity, impact, and type (e.g., data breach, malware infection, denial-of-service attack)
- PC7.** take immediate containment measures, such as isolating affected systems or disabling compromised accounts
- PC8.** perform thorough investigations to understand the root cause of attacks, identify attacker methods, and assess damage extent

Incident Containment, Mitigation, and Recovery

To be competent, the user/individual on the job must be able to:

- PC9.** develop a detailed timeline of events to reconstruct the attack sequence
- PC10.** isolate affected systems or networks to prevent further spread of the attack
- PC11.** identify and remove malicious software from infected systems effectively
- PC12.** patch known vulnerabilities to prevent future attacks
- PC13.** use backup procedures to restore lost or corrupted data
- PC14.** document the incident response process for future reference

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** The functionality and application of advanced security tools such as Wireshark, KisMAC, and NetStumbler for network traffic monitoring, system log analysis, and detecting suspicious activities.
- KU2.** The role of threat intelligence feeds in identifying emerging threats, attack patterns, and potential security vulnerabilities.
- KU3.** The working principles of Security Information and Event Management (SIEM) solutions, including how to aggregate, correlate, and analyze security events from multiple sources.
- KU4.** Various incident categorization techniques, including how to assess security incidents based on severity, type, and impact (e.g., malware infection, data breach, denial-of-service attacks).
- KU5.** Different methods of containment and mitigation, such as isolating affected systems, disabling compromised accounts, and preventing further attack propagation.
- KU6.** Root cause analysis (RCA) methodologies to investigate security incidents, reconstruct attack sequences, and assess attacker techniques.
- KU7.** Techniques for patching vulnerabilities, removing malicious software, and restoring lost or corrupted data using backup procedures to ensure business continuity.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Utilize network monitoring and intrusion detection tools to continuously analyze user behavior, detect unauthorized access attempts, and identify anomalies in system logs.
- GS2.** Apply behavioral analysis algorithms to detect deviations from normal network traffic patterns and system operations.
- GS3.** Correlate security alerts from multiple sources to accurately determine potential incidents and take necessary countermeasures.
- GS4.** Perform forensic investigations to reconstruct attack timelines, identify attack vectors, and analyze the extent of security breaches.
- GS5.** Maintain detailed documentation of the incident response process, including attack detection, containment, mitigation steps, and lessons learned for future reference.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Threat Detection, Incident Analysis, and Response</i>	18	28	-	11
PC1. utilize advanced security tools (Wireshark, KisMAC, NetStumbler, etc.) to monitor network traffic, system logs, and user behavior continuously for signs of suspicious activity	3	4	-	1
PC2. leverage threat intelligence feeds to remain informed about emerging threats and attack patterns	2	3	-	1
PC3. employ Security Information and Event Management (SIEM) solutions to aggregate and correlate security events from multiple sources to identify potential incidents	2	4	-	2
PC4. apply algorithms for detecting deviations from normal behavior, including unusual network traffic or unauthorized access attempts	2	3	-	2
PC5. correlate alerts from various sources when suspicious activities are detected to determine potential incidents	2	4	-	2
PC6. categorize incidents based on severity, impact, and type (e.g., data breach, malware infection, denial-of-service attack)	3	3	-	1
PC7. take immediate containment measures, such as isolating affected systems or disabling compromised accounts	2	4	-	1
PC8. perform thorough investigations to understand the root cause of attacks, identify attacker methods, and assess damage extent	2	3	-	1
<i>Incident Containment, Mitigation, and Recovery</i>	12	22	-	9
PC9. develop a detailed timeline of events to reconstruct the attack sequence	2	4	-	2
PC10. isolate affected systems or networks to prevent further spread of the attack	2	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. identify and remove malicious software from infected systems effectively	2	4	-	2
PC12. patch known vulnerabilities to prevent future attacks	2	3	-	2
PC13. use backup procedures to restore lost or corrupted data	2	4	-	1
PC14. document the incident response process for future reference	2	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N0963
NOS Name	Identify and response to cyber attack
Sector	IT-ITeS
Sub-Sector	
Occupation	Information and Cyber Security
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.
7. Candidate needs to achieve 70% or more for successfully passing the QP.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N0961.Detect threats and monitor events using Sysinternal Tools	30	50	0	20	100	29
SSC/N0962.Analyze security events and response to detected threats	30	50	0	20	100	29
SSC/N0963.Identify and response to cyber attack	30	50	0	20	100	29
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	13
Total	110	180	-	60	350	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.